

Fig. 3: Actual-size PCB pattern of LME49710 audio amplifier

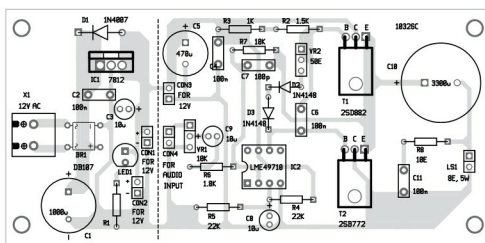


Fig. 4: Component layout of the PCB

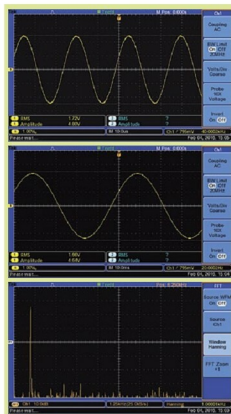


Fig. 5: Audio output waveforms

capacitor C11 form a low-impedance path for the high-frequency noise that appears across speaker LS1.

Duplicate the same circuit for another channel to make it a stereo amplifier. The same power supply at CON2 may be used for the second amplifier unit.

Construction and testing

An actual-size, single-side PCB for LME49710 audio amplifier is shown in Fig. 3 and its component layout in Fig. 4. Power supply and amplifier units are separated by dotted lines where right side of the PCB is for the audio amplifier unit. A similar PCB can be used for the second amplifier unit.

Fit VR1 and CON4 on the front side of the panel for volume control and audio input connections, respectively. Adjust VR2 to around 11 ohms. We have used 2-pin connector for audio input but you can use RCA

PARTS LIST

Semiconductors:

- IC1 - LM7812, 12V voltage regulator
- IC2 - LME49710 op-amp
- BR1 - DB107 bridge rectifier
- T1 - 2SD882 npn transistor
- T2 - 2SB772 pnp transistor
- LED1 - 5mm LED
- D1 - 1N4007 rectifier diode
- D2, D3 - 1N4148 signal diode

Resistors (all 1/4-watt, 45% carbon):

- R1 - 2.2-kilo-ohm
- R2 - 1.5-kilo-ohm
- R3 - 1-kilo-ohm
- R4, R5 - 22-kilo-ohm
- R6 - 1.8-kilo-ohm
- R7 - 10-kilo-ohm
- R8 - 10-ohm
- VR1 - 10-kilo-ohm potentiometer
- VR2 - 50-ohm potentiometer

Capacitors:

- C1 - 1000µF, 35V electrolytic
- C2, C4, C6, C11 - 100nF ceramic disk
- C3, C8, C9 - 10µF, 25V electrolytic
- C5 - 470µF, 25V electrolytic
- C7 - 100pF ceramic
- C10 - 3300µF, 25V electrolytic

Miscellaneous:

- LS1 - 8-ohm, 5W speaker
- X1 - 230V AC primary to 15V, 1A secondary transformer
- CON1-CON4 - 2-pin connector terminal
- CON5 - 2-pin connector terminal

female connector or stereo female socket for audio input and connect the music player/iPod/mobile. Enclose the setup in a small plastic box. Connect the transformer primary to 230V AC mains supply.

Fig. 5 shows the audio output waveforms seen in an oscilloscope with audio input taken from the function generator. The waveforms you see are measured across the load of 7.5 ohms.

Results.

Output impedance: 0.01-ohm

Frequency response: flat from 20Hz - 40kHz

Total harmonic distortion: 1.2% at 3W peak

Test frequency: 1kHz sine of 1Vpp input

Can easily drive 3-ohm to 8-ohm speakers **EFY**



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